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Lower Jurassic magnetostratigraphy at the Breggia Gorge (Ticino, Switzerland) and Alpe Turati (Como, Italy)

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Summary. The Breggia gorge (Ticino, Switzerland) provides a continuous outcrop of Mesozoic limestones covering the time between the Lower Jurassic and Upper Cretaceous. The Jurassic section from the Lower Pliensbachian to Aalenian is well dated by ammonites. The limestones of these stages are a good quality recorder of the polarity of the geomagnetic field during this time interval. More than 500 cores have been drilled along a 120 m thick section with a median distance between cores of about 20 cm. Detailed thermal and af demagnetization and further rock magnetic studies were used to determine the characteristic NRM direction for each sample. Initial NRM intensities and low field susceptibilities reflect variations in oxidation potential and sedimentation rate during carbonate deposition. The Breggia section covers an estimated 12 Myr of Lower to Middle Jurassic time during which more than 60 distinct reversals of the geomagnetic field are recorded. This corresponds to a comparatively high mean reversal frequency of about 5 reversals Myr^{-1}. The Breggia polarity zonation can be correlated with the nearby section at Alpe Turati (Como, Italy) and comparable polarity patterns are recognized in other biostratigraphically well-dated magnetostratigraphic profiles from Hungary and Umbria (Italy).

Introduction

The development of cryogenic and digital flux-gate magnetometers, which allow rapid and accurate measurement of weakly magnetized rock sequences such as limestones or deep-sea sediments, has resulted in a proliferation of magnetostratigraphic investigations. Many of these studies have concentrated on Late Cretaceous or Palaeogene strata (e.g. Lowrie & Alvarez 1977; LaBrecque *et al.* 1981; Lowrie *et al.* 1982) and excellent correlation between magnetostratigraphic polarity intervals and Cainozoic marine magnetic anomalies has been obtained. However, the correlation of the Late Jurassic to Early Cretaceous 'M-sequence' of marine anomalies (Larson & Pitman 1972) and the significance and duration of the Jurassic magnetic quiet interval is not so clear-cut. The dating of the beginning and end of the M-sequence of oceanic anomalies is still imprecise. Larson & Hilde (1975) estimated the age of the oldest M-anomalies at about 153 Myr on the basis of basal sediment ages in

DSDP holes. The lack of prominent oceanic magnetic lineations older than this time limit defines a Jurassic magnetic quiet zone which has been variously interpreted as being due to: (1) a period of constant (normal) magnetic polarity, (2) a period of constant (reversed) magnetic polarity (Steiner 1980), (3) a period of high reversal frequency but low dipole moment (Cande, Larson & LaBrecque 1978), or (4) extensive viscous remagnetization (Lowrie & Kent 1978).

Detailed palaeomagnetic studies of continuously deposited Jurassic sections now exposed on land may establish the nature of the Jurassic quiet zone and also set time constraints on its duration. McElhinny & Burek (1971) concluded on the basis of the available data that normal geomagnetic polarity predominated during the Jurassic with a mixed polarity interval starting at about the Oxfordian/Kimmeridgian boundary. Although this picture was not very accurate because of the poor stratigraphic control of the data used, it did not change appreciably for some time (Creer 1971; Johnson, Nairn & Peterson 1972; Pechersky & Khramov 1973; Irving & Pullaiah 1976).

It is usually difficult to obtain accurate age control in continental deposits, but high quality biostratigraphic zonation is often found in marine limestone sequences. Polarity patterns derived from Late Jurassic limestone sequences in the Southern Alps (Ogg 1981; Channell, Ogg & Lowrie 1982a) suggest that the oldest of the M-sequence anomalies (M-25) has an age close to the Oxfordian/Kimmeridgian boundary. Oxfordian limestones in Franconia (Southern Germany) are normally magnetized (Heller 1977). A Lower Jurassic Pliensbachian pelagic limestone section from Hungary (Márton, Márton & Heller 1980)

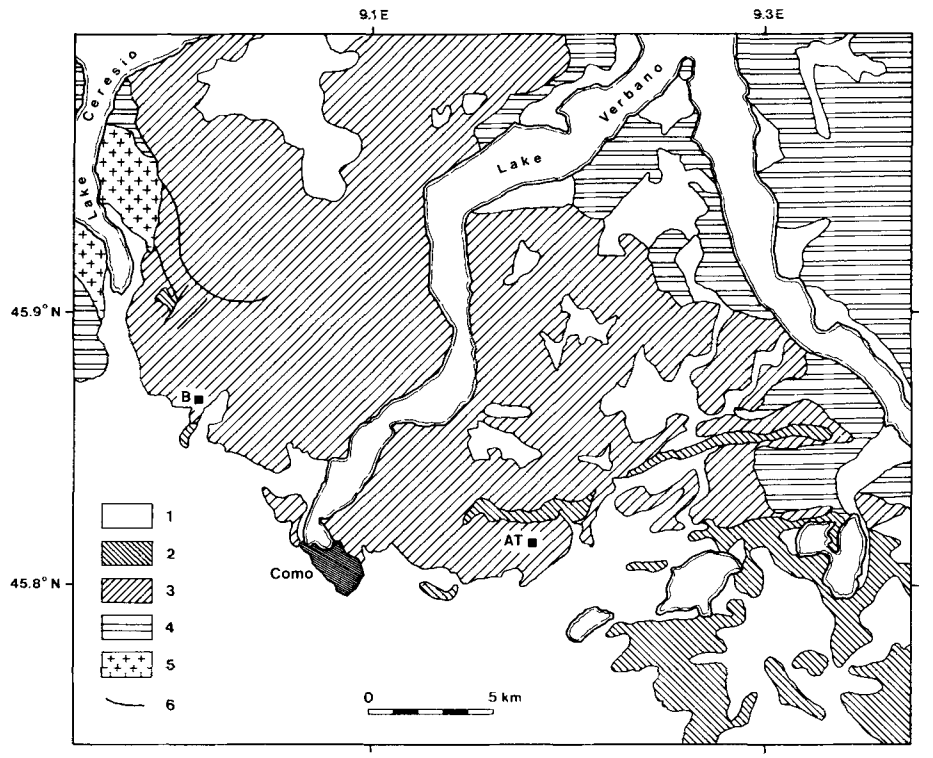


Figure 1. Geological sketch map of the Western Lombardy area (Southern Alps), with sampling localities indicated (B: Breggia; AT: Alpe Turati). Key: (1) Quaternary and Tertiary deposits, (2) Cretaceous sediments, (3) Jurassic sediments, (4) Triassic sediments, (5) Permian volcanics, (6) Major faults.

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shows frequent reversals of natural remanent magnetization. Mixed polarities also have been found (Channell *et al.* 1982a) in highly condensed Lower Jurassic limestone sections (Sinemurian-Toarcian) from Umbria (Italy) and Bathonian limestone sections from the Southern Alps (Ogg 1981). The Oxfordian and Late Middle Jurassic limestones from Umbria have normal magnetization directions. Therefore Channell *et al.* (1982a) restrict the Jurassic quiet zone tentatively to the Callovian and Oxfordian stages only.

In this paper we present a continuous polarity record for the Pliensbachian to Bajocian stages derived from limestone exposures in the Breggia gorge (Ticino, Switzerland) and at Alpe Turati (Como, Italy), in the Southern Alps (Fig. 1).

Geological setting

The relief of the Southern Alps was levelled flat towards the end of the Permian; during the Early Triassic marine conditions began to prevail. In the Middle Triassic the south-alpine epicontinental region was differentiated into blocks in which subsidence was balanced out by shallow-water carbonate sedimentation (Bernoulli 1980). With the opening of the Tethys ocean in the Early Jurassic the region, being part of the southern continental margin, was affected by synsedimentary normal faulting. The previous Upper Triassic shallow-water carbonate platform was segmented into smaller blocks. A complex pattern of submarine swells and basins (Gaetani 1975) was generated which exerted control on the Jurassic sedimentation. Subsidence and sedimentation rates decreased and were more evenly distributed over the south-alpine margin. Submergence of the increasingly starved distal continental margin resulted in the deposition of a pelagic carbonate sequence. The facies was mainly determined by continuous synsedimentary faulting and increasing water depth, which is reflected in the solution facies of the pelagic carbonates (Bernoulli 1980).

The Breggia profile has attracted the attention of geologists for a long time because it is a unique place in which to study the continuous sedimentary record of the Lombardian sequence deposited in the Generoso basin through Jurassic to Cretaceous times. Fig. 2 shows the Lower Jurassic lithostratigraphy and biostratigraphy of the Breggia section according to Laubscher & Bernoulli (1980) and Wiedenmayer (1980). For the middle and lower part of the section Wiedenmayer (1980) has worked out all the details of palaeontological division into ammonite zones and subzones. For the upper part we use the results given by Renz (1920), Donovan (1958), Bernoulli (1964), Bernoulli & Wiedenmayer (1967) and Laubscher & Bernoulli (1980).

In the lower part of the section (Fig. 2) the Lombardian Siliceous Limestone formation (Hettangian to basal Domerian) is formed by blueish-grey, well-bedded, spongolithic siliceous calcilutites with bands and nodules of replacement chert (Laubscher & Bernoulli 1980). Up-section the formation grades into the Molino member which comprises to Davoei zone of the Upper Carixian and the Stokesi subzone of the Lower Domerian Margaritatus zone. The light grey, hemipelagic limestones of the Molino member are interbedded with silty micaceous marls, the bands and nodules of granular chert being reduced. The Morbio member, extending through the remainder of the Domerian and including the Lower Toarcian Tenuicostatum zone represents the base of the Rosso Ammonitico Lombardo formation. It consists of an alternation of red pelagic limestones and marls which grade into the nodular marl and limestone facies of the upper Rosso Ammonitico Lombardo. In the overlying white and red limestones and marls with pelagic bivalves (Upper Toarcian to Lower Bajocian) ammonites become less abundant and have been found in redeposited strata only. Part of the section is repeated in a large, approximately 9 m thick submarine slump, overlain by pelagic and turbiditic siliceous limestones.

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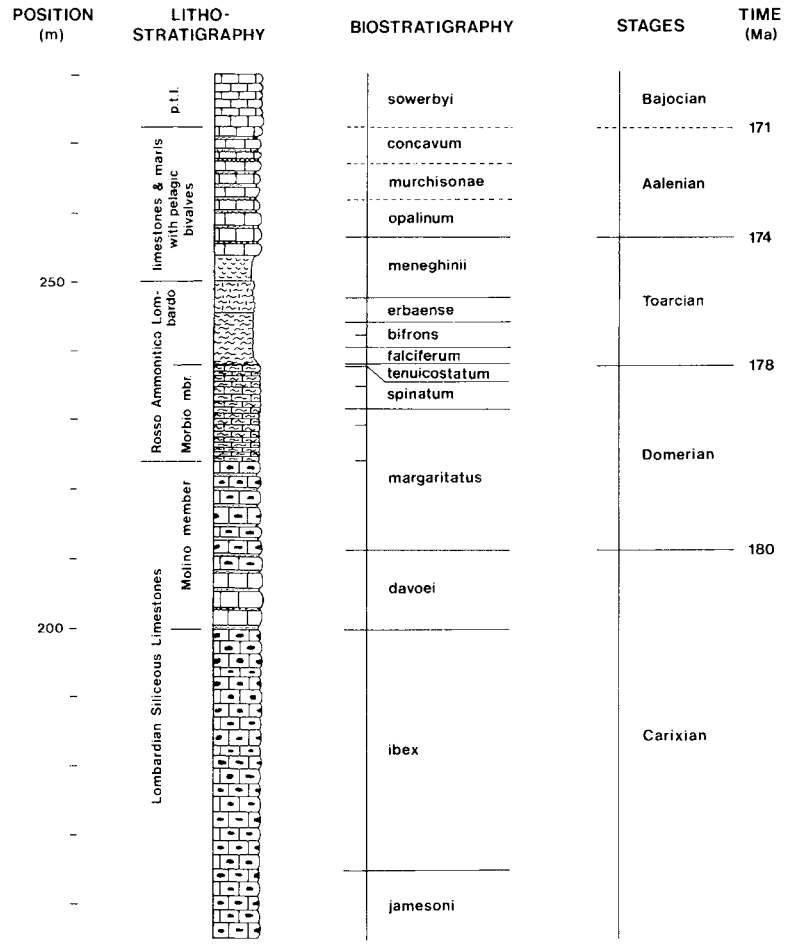


Figure 2. Litho- and biostratigraphy (Wiedenmayer 1980) of the Breggia profile, representing the Carixian to Bajocian part of the Lombardian sequence. Absolute ages of stage boundaries according to Van Hinte (1976).

The Breggia section has a more or less uniform 50–60° dip south-westwards on the southern flank of the Monte Generoso anticline. The Breggia stream cuts through the tilted beds perpendicular to the strike and provides a very fresh outcrop. More than 500 oriented samples (volume = 11.4 cm^3) have been collected across the approximately 120 m thick section by means of a gasoline powered portable drill with a median distance between cores of about 20 cm.

The 10 m thick section sampled at Alpe Turati dips very gently south-westwards at about 5° and comprises the 'calcari rosso-vino a spinatum e lariense' (Venzo 1952; Gaetani & Fantini-Sestini 1978). These limestones correspond litho- and biostratigraphically (Wiedenmayer 1980) to the Morbio member of the Breggia section.

Laboratory techniques

The remanent magnetization of the samples was measured using a three-axis cryogenic magnetometer interfaced to a minicomputer, which allows for the integration of a pre-selected number of readings (in general 10) per measurement. A sensitivity limit of

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6.0×10^{-6} A m^{-1} is achieved. Replicate measurements of each sample in three or six different positions relative to the magnetometer axes overdefine the remanence vector, so that errors of measurement, whether instrumental or due to sample instability or inhomogeneity may be recognized at each stage of sample treatment (Lowrie, Channell & Heller 1980).

In the course of an intensive pilot study a large number of samples was demagnetized progressively in order to separate individual components of natural remanent magnetization (NRM). Wherever possible, when two samples from the same core were available, one was subjected to alternating field (af) demagnetization and the other to thermal demagnetization. Af cleaning with peak alternating fields up to 60 mT was carried out using a two-axis tumbler within which the Earth's magnetic field was cancelled by a Helmholtz coil system. Prior to af demagnetization the samples were always heated to 150°C in order to remove possible magnetization components residing in goethite (Heller 1978) which would be most probably of secondary origin. Thermal demagnetization up to 700°C (if necessary) was performed in air in a furnace contained in μ-metal shielding. After each step of thermal treatment, bulk magnetic susceptibility was monitored in order to recognize any temperature-dependent changes in magnetic mineralogy. Remagnetization of the samples by the Earth's magnetic field was kept at a minimum by transporting the samples from the demagnetization device to the magnetometer in μ-metal shields and by having the magnetometer surrounded by large field-compensating Helmholtz coils. The components analysis of NRM was facilitated by the use of vector diagrams (Zijderveld 1967) and three-dimensional least squares fitting (Kirschvink 1980). The identification of magnetic minerals followed the isothermal remanent magnetization methods applied to limestones by Lowrie & Heller (1982).

Properties of remanent magnetization

ISOTHERMAL REMANENT MAGNETIZATION (IRM)

The coercivity spectrum of a 'white-coloured' sample was investigated by giving it an isothermal remanent magnetization (IRM) in progressively increasing DC fields up to 3.0 T by the use of a cryogenic magnet (Fig. 3a). The IRM acquisition curve of this sample indicates the predominance of a lower coercivity mineral – probably magnetite – as the carrier of the remanence. More than 98 per cent of IRM is acquired below 0.3 T, the maximum coercivity for magnetite. Progressive thermal demagnetization of the saturation IRM (Fig. 3b) gives maximum unblocking temperatures around 550°C in support of the presence of magnetite in the 'white' limestones. Magnetite in limestones is generally assumed to be of detrital origin and to carry a magnetization which is almost synchronous with the deposition of the carbonates (Lowrie & Heller 1982).

The IRM of the 'red' limestones is characterized by a broad range of coercivities (Fig. 3c), saturation not being achieved even in a field of 4.0 T. The inflexion of the IRM thermal demagnetization curve at 100°C (Fig. 3d) and the fact that saturation is not often reached at the highest fields available, point towards the presence of goethite; a similar inflexion at 550°C indicates a minor contribution from magnetite. Above 550°C the IRM still holds about 30 per cent of the room temperature intensity due to a strong haematite contribution. The haematite is finally unblocked at 650°C.

The IRM acquisition curve of a sample with mixed magnetic mineralogy including magnetite, haematite and goethite (Fig. 3e) is not much different from that of a 'red' sample (Fig. 3c). Due to the presence of more magnetite the increase in magnetization below 0.3 T is slightly stronger, and at high fields again saturation is not achieved. In order to separate the IRM contributions of the three minerals more clearly, a second IRM in an antiparallel field of 0.2 T was given to the sample prior to heating. In this case the decreasing

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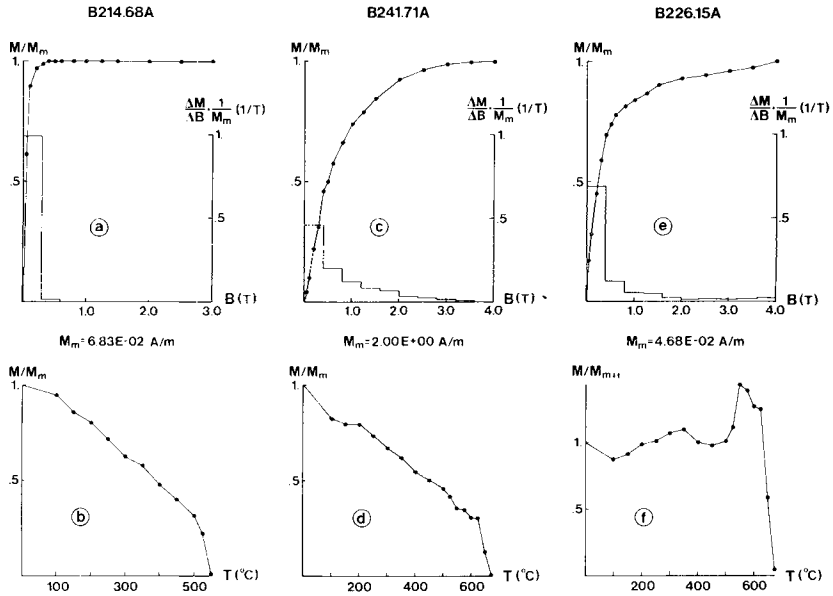


Figure 3. Acquisition and thermal demagnetization of isothermal remanent magnetization (IRM). (a, b) 'White' limestone from the upper part of the Lombardian Siliceous Limestones. (c, d) 'Red' marl from the Lombardian Ammonitic Rosso. (e) IRM acquisition of a 'red' limestone from the Morbio member containing magnetite, haematite and goethite. (f) Progressive thermal demagnetization of an IRM being composed of a 4 T component and an antiparallel 0.2 T component in order to separate high and low coercivity magnetizations during heating.

intensities upon heating (Fig. 3f) point to the removal of mainly high coercivity components, whereas increasing parts of the curve reflect demagnetization of mainly low coercivity material. Up to 100°C goethite is demagnetized. Maximum unblocking temperatures of magnetite reach 550°C. Between 350 and 450°C a component with relatively low blocking temperatures, but high coercivities – probably pigmentary haematite – is unblocked, but again the major IRM contribution from haematite is lost between 550 and 650°C.

NATURAL REMANENT MAGNETIZATION (NRM)

Fig. 4(a–d) show palaeomagnetic results from two samples from the upper part of the Lombardian Siliceous Limestones. These samples characterize the magnetic behaviour of the light to dark grey coloured limestone of the upper part of the Breggia section.

Sample B212.37B was demagnetized in alternating fields. Prior to af cleaning the sample was heated to 150°C (first demagnetization step in Fig. 4b). This step removes a component of normal polarity (Fig. 4a). During further af treatment the magnetization vector decays univectorially towards the origin of the projection which is closely approached by 60 mT. A stable remanence component, the characteristic remanent magnetization (ChRM), is easily determined in the region between 20 and 50 mT.

Thermal cleaning of the 'white' limestones often resulted in irregular demagnetization curves at higher temperatures. The remanence vector underwent major changes as a result of heating to temperatures above 500°C. The NRM of the thermally demagnetized sample B204.52B shows erratic changes in direction (Fig. 4c) and intensity (Fig. 4d) above 550°C. In addition the low field susceptibility of the sample increases dramatically during heating to temperatures above 400°C. This change is attributed to the destruction of clay minerals

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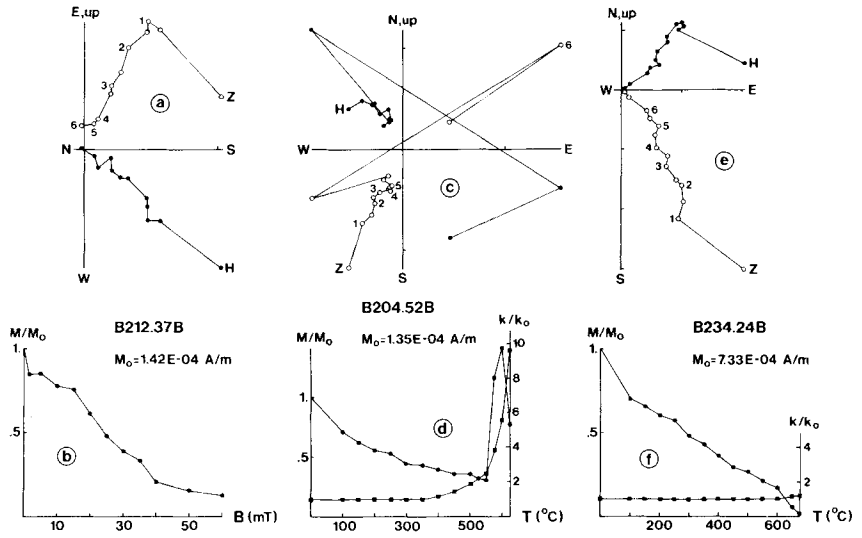


Figure 4. NRM-intensity and -vector behaviour of Lower Jurassic limestones from the Breggia section during demagnetization. (a, b) Alternating field demagnetization of a 'white' limestone. (c, d) Thermal treatment of a 'white' limestone sample. (e, f) Thermal demagnetization of a 'red' limestone sample. Open and solid dots on vector diagrams indicate components in the vertical (Z) and horizontal (H) planes, respectively. Numbers along vector paths refer to either peak AC demagnetization fields in tens of mT or temperatures in hundreds of °C. Squares on intensity plots denote relative susceptibility as measured after each heating step. Unit on vector plots (a) and (c) 1×10^{-4} A m^{-1}, on vector plot (e) 2×10^{-4} A m^{-1}.

at higher temperatures, and not to oxidation of goethite, for which there is no indication in the IRM acquisition curve (Fig. 3a). The mineralogical changes at elevated temperatures lead to the formation of a new magnetic mineral with high susceptibility – probably magnetite – which shows strong viscous behaviour above 550°C probably due to very small grain size.

For the 'red-coloured' limestones and marls which constitute the middle and upper part of the Breggia profile and the whole Alpe Turati section, thermal demagnetization proved to be the optimum cleaning procedure. The large change in direction (Fig. 4e) and intensity (Fig. 4f) observed after the first heating step is diagnostic of a rather strong goethite contribution (compare red carbonate IRM in Fig. 3c–f) to the NRM of this sample. Additional components are removed up to 250 and 500°C, which probably reside in very fine grained (pigmentary) haematite (*cf.* IRM thermal demagnetization in Fig. 3f). Above 500°C the remanence vector decays univectorially and monotonously towards the origin of projection yielding a well defined characteristic remanence direction. This component is removed at 650°C.

SUMMARY OF MAGNETIC PROPERTIES

The limestones often contain all three positively identified magnetic minerals in different proportions. In addition to magnetite as a principal magnetic constituent, the 'white' limestones often contain minor amounts of goethite. In the 'red' limestones usually variable proportions of haematite, magnetite and goethite control the magnetic properties. It was found, however, that the direction of the haematite NRM component does not differ significantly from that of the magnetite component (Fig. 4e). This suggests that both components were acquired within a comparatively short time of each other. Channell *et al.* (1982b) have shown that haematite in Palaeocene Scaglia Rossa limestones from Umbria was formed

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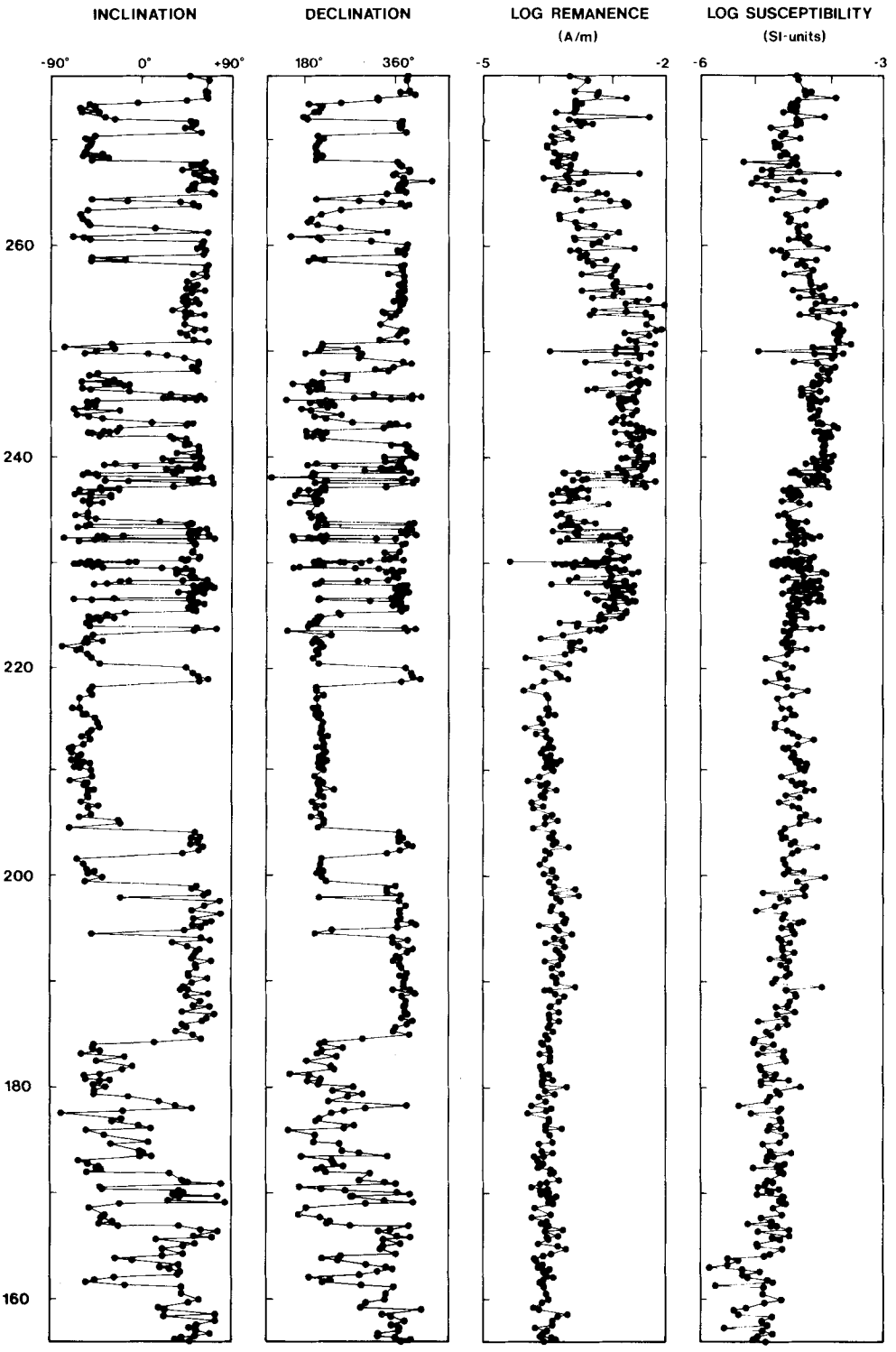


Figure 5. Inclination and declination of the stable characteristic NRM components, initial NRM intensity and low field susceptibility versus profile depth [m] along the Breggia section.

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diagenetically nearly contemporaneously with the deposition of the sediment. On the other hand, high coercivity goethite (probably produced during weathering) carried secondary NRM components (Heller 1978). Because of the very low unblocking temperatures these components are easily removed by thermal cleaning well below 150°C.

Magnetic stratigraphy

Declination and inclination of the characteristic remanent magnetization as well as low field susceptibility and initial NRM intensity have been plotted as functions of stratigraphic position for the Breggia section (Fig. 5). Positive and negative inclinations with corresponding northerly and southerly declinations constitute a pattern of normal and reversed polarity zones. Both polarities occur in about equal proportions (Table 1). The mean inclination is unexpectedly steep in the Breggia section, which during the Lower Jurassic was located on the southern continental margin of the Tethys. It also differs significantly from the mean value in the Alpe Turati section (Table 1), which is less disturbed tectonically and, as far as palaeomagnetic directions are concerned, more representative for the southern alpine region (VandenBerg & Zijderveld 1982). The tectonic displacement at the Breggia gorge, consisting of a large drag fault (Bernoulli 1964), obviously cannot be corrected for by a simple bedding rotation about a horizontal axis. Nevertheless this correction is adequate to allow recognition of the succession of polarity zones.

Two factors, namely deposition rate and oxidation potential, control the intensity of NRM and susceptibility. Using Van Hinte's (1976) age estimates for the Lower Jurassic stage boundaries we can calculate average sedimentation rates for the Carixian to Aalenian stages at the Breggia gorge. The biostratigraphic stage boundaries (Wiedenmayer 1980) are very well defined for the lower and middle part of the profile (Fig. 2), whereas the Aalenian and Bajocian ammonite zones are only vaguely determined. The deposition rate decreases steadily by about 1 order of magnitude from Carixian (23 m Myr^{-1}) to Aalenian (1.8 m Myr^{-1}). Admitting these figures to be in error to some extent, because the estimates of the stage boundaries are largely intuitive, the trend nevertheless is considered to be significant. The changing sedimentation rate is also reflected in a gradual, nearly linear increase of susceptibility by about 2 orders of magnitude from the bottom to the top of the lower 90 m of the Breggia profile (Fig. 5). The input of detrital magnetite into the pelagic realm is assumed to be more or less constant over time. Therefore its relative contribution to susceptibility increases with decreasing deposition rate. Between 220 and 225 m, where the lithology changes from the Lombardian Siliceous Limestones facies to Rosso Ammonitico Lombardo facies, the NRM intensity changes rather suddenly by an order of

Table 1. Mean directions of characteristic remanent magnetization.

Section	N	n	Dec. (Geographic)	Inc. (Geographic)	Dec. (Tilt corr.)	Inc. (Tilt corr.)	α_{95}	Polarity (per cent) N	R
Breggia	538	642	27.2	6.3	14.1	52.7	2.2	48	52
Alpe Turati	62	62	333.6	46.4	337.3	43.2	6.7	34	66

N = number of cores.
n = number of specimens measured.
Dec., Inc. = declination and inclination before (geographic) and after (tilt corr.) bedding correction.
α_{95} = Fisher's (1953) confidence angle.
Polarity column indicates percentage of normally (N) and reversely (R) polarized samples respectively.

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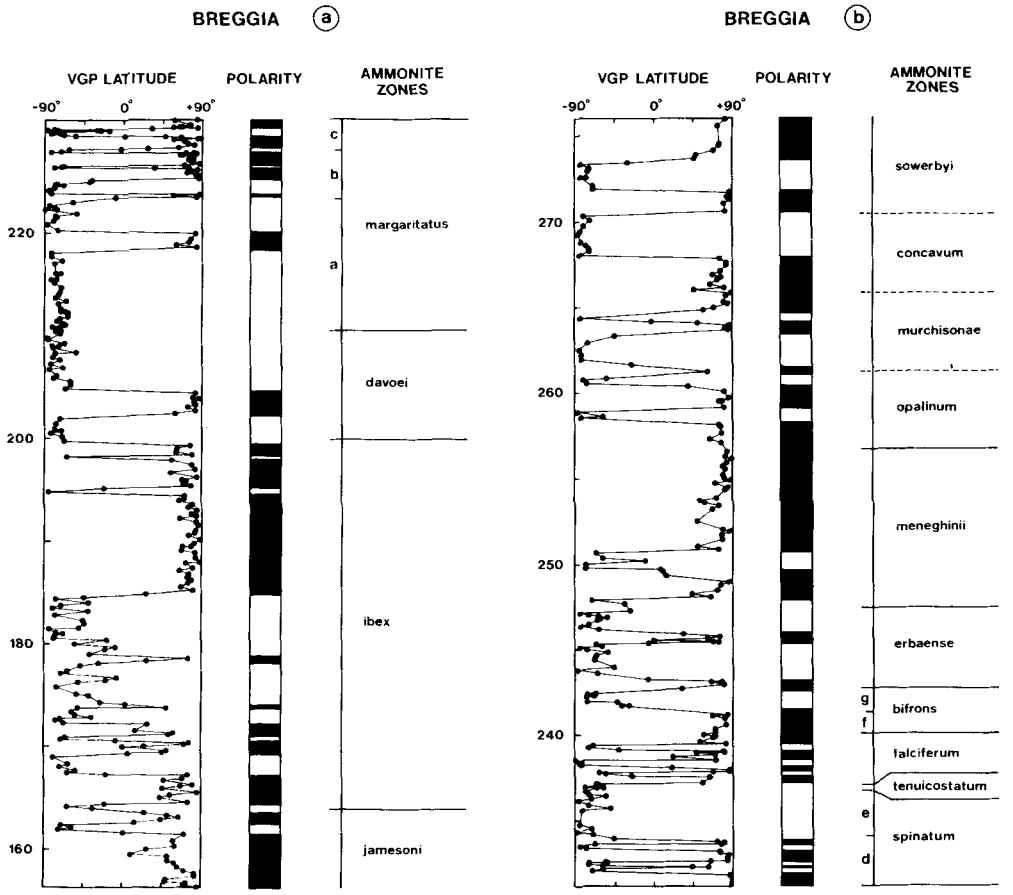


Figure 6. VGP latitudes, magnetic polarity and ammonite zonation for (a) the lower part, (b) the upper part of the Breggia section on a metre scale. Ammonite subzones according to Wiedenmayer (1980): (a) stokesi, (b) subnodosus, (c) gibbosus, (d) apyrenum, (e) hawskerense, (f) semipolitum, (g) sublevisoni. Normal polarity zones are black, reversed zones are white.

magnitude. The formation of haematite in the red to purple coloured limestones gives rise to this prominent increase in remanence intensity.

In magnetostratigraphic studies the polarity zones may be determined from plots of virtual geomagnetic polar (VGP) latitudes which are computed from declination and inclination of the characteristic remanence for each sample (*cf.* Fig. 5) with respect to the section's palaeomagnetic mean pole. Using this convenient technique, polarity sequences have been deduced from the distribution of VGP latitudes for the sections at the Breggia gorge (Fig. 6) and Alpe Turati (Fig. 7).

The polarity zonation of the lower part of the Breggia profile (Fig. 6a; 155–180 m) is not very clearly developed. This is mainly due to the very low intensity of the characteristic NRM (less than 1×10^{-5} A m^{-1}). Above 180 m profile depth (Fig. 6a, b) distinct polarity zones are observed. Their apparent duration is highly variable and in several cases documented by only one or two successive cores. The transition from one polarity to the other always occurs within a few centimetres representing a time interval of the order of 10 000 yr so that very few intermediate remanence directions or VGP positions have been detected. Sometimes antiparallel NRM directions have been observed even within single cores, which – drilled perpendicular to bedding – cut a polarity boundary.

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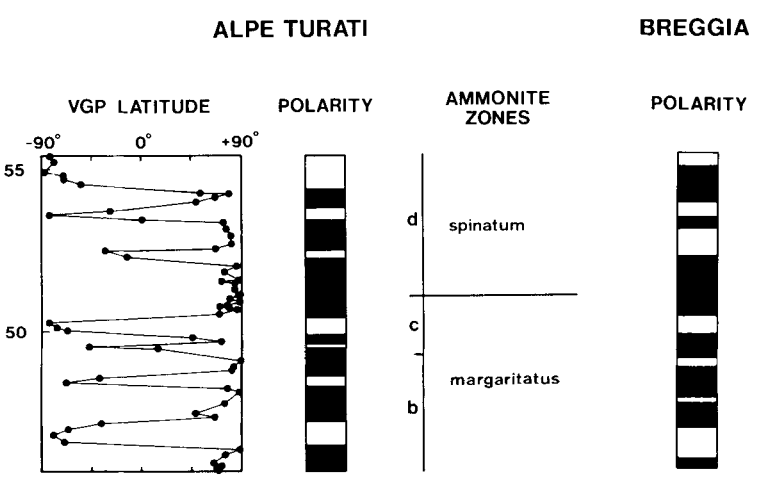


Figure 7. VGP latitudes, magnetic polarity and ammonite zonation for the Alpe Turati section. Polarity distribution compared to that of the contemporaneous part of the Breggia section. Ammonite subzones as in Fig. 6.

The Breggia profile includes 67 normal and reversed polarity zones covering the Carixian to Bajocian time interval (Fig. 6). By reference to Van Hinte's (1976) time-scale this results in a high average reversal rate for the covered time span of about 5.5 reversals Myr^{-1}.

The correlation of polarity distribution between the Breggia section and the short Alpe Turati section (Fig. 7) is good. The main polarity features are similar on both profiles, but the apparent duration of the intervals differs slightly. Although the biostratigraphy of both profiles is known as accurately as possible, the sedimentation record (especially in this red, condensed limestone facies) is probably neither continuous nor regular giving rise to the observed duration differences of magnetic polarity zones. This is taken as a warning that without detailed biostratigraphic control magnetostratigraphic correlation of sedimentary sequences is extremely difficult to achieve if the reversal rate is high in comparison to the sedimentation rate. The palaeontological control is especially important when polarity records of such sequences in different basins are correlated.

The polarity sequences of the Breggia (B) and Alpe Turati (AT) sections have been redrawn on a linear time-scale in Fig. 8 together with the polarity records preserved in other contemporaneous carbonate deposits from Umbria (Channell *et al.* 1982a) and Hungary (Márton *et al.* 1980). Although the sections originate from different sedimentary basins with different deposition history, it is possible to recognize some common magnetozone features. The base of the Valdorbia (VD) profile can be tied biostratigraphically to the Breggia section so that the same succession of polarity zones although not exactly in phase can be followed up in both data sets. The correlation implies that the upper part of the Valdorbia section extends in age into the Lower Aalenian. In the Carixian and Domerian the sedimentation rate in the Southern Alps is very high. The thickness of the Pliensbachian in the Breggia amounts to about 80 m, whereas the Hungarian Pliensbachian at Bakonycsernye (BC) is only about 9 m thick. Thus the large number of short frequency polarity zones observed in the Breggia and Alpe Turati sections constitutes packages of preferred normal or reversed field episodes which may be correlated to the simpler polarity zonation of the profiles at Cingoli (CI) and Bakonycsernye where the sedimentation rate was much more reduced. In the Pliensbachian three major magnetozones seem to be recorded: a reversed zone in the upper Carixian, an essentially normal zone in the Middle Domerian and a reversed zone across the Domerian–Toarcian boundary. These marker events seem to allow

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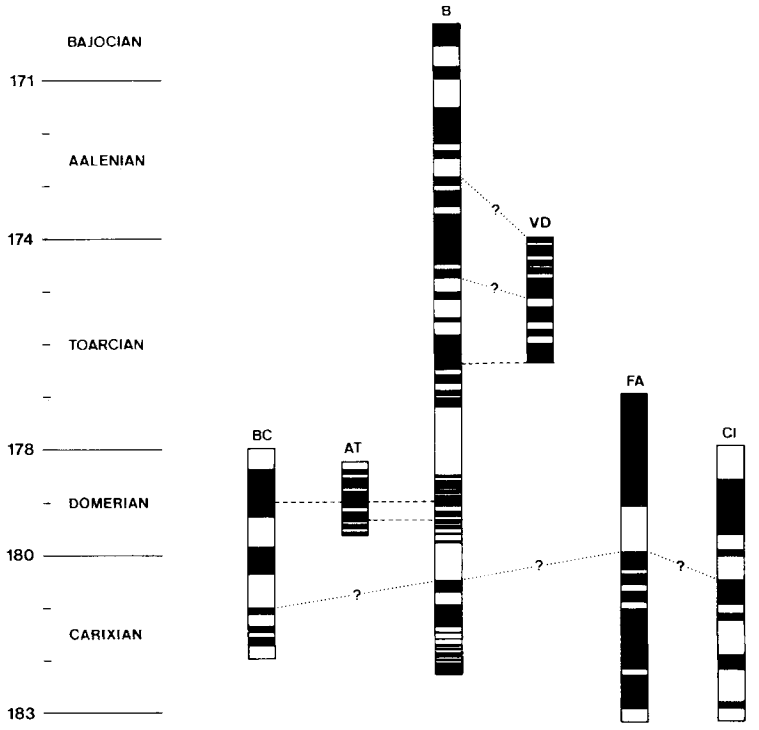


Figure 8. Geomagnetic polarity as recorded by six Lower to Middle Jurassic magnetostratigraphic sections [CI (Cingoli), FA (Fonte Avellana), VD (Valdorbia) after Channell *et al.* (1982a); BC (Baconycsernye) from Márton *et al.* (1980); AT (Alpe Turati) and B (Breggia), this work]. Normal magnetozones are black, reversed white. The dashed tie-lines are well established palaeontological zone boundaries, the dotted tie-lines with question marks represent possible magnetostratigraphic correlation horizons. Linear time-scale. Absolute ages (Myr) correspond to the time-scale of van Hinte (1976).

for a possible correlation even with the very poorly dated section at Fonte Avellana, which probably due to an unrecognized high sedimentation rate does not include Toarcian sediments so that the time interval represented by this section is much shorter than indicated in Fig. 8.

Conclusions

The Lower to Middle Jurassic limestones exposed in the Breggia gorge contain a clearly defined polarity record which shows that the geomagnetic field changed polarity frequently during this time interval. There is no comparable record of marine magnetic anomalies because no seafloor of this age has yet been identified. However, a clear and distinct polarity pattern has been evaluated from the analysis of the characteristic NRM directions. During the Pliensbachian the reversal rate was extremely high with more than 13 reversals Myr^{-1}. From Toarcian to Bajocian the reversal rate comes down to similar values as those for the most recent polarity epochs. High reversal rates may be resolved only with high sedimentation rates and a clear magnetostratigraphic correlation between sections seems to be possible only if detailed biostratigraphic control is available.

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